j make a salt and calculate the percentage yield of product, e.g. preparation of a double salt (ammonium iron(II) sulfate from iron, ammonia and sulfuric acid)

k carry out and interpret the results of simple test tube reactions, such as displacements, reactions of acids, precipitations, to relate the observations to the state symbols used in equations and to practise writing full and ionic equations.

## 1.4 Energetics

### Students will be assessed on their ability to:

a demonstrate an understanding of the term *enthalpy change*, $\Delta H$

b construct simple enthalpy level diagrams showing the enthalpy change

c recall the sign of $\Delta H$ for exothermic and endothermic reactions, e.g. illustrated by the use of exo- and endothermic reactions in hot and cold packs

d recall the definition of standard enthalpy changes of reaction, formation, combustion, neutralisation and atomisation and use experimental data to calculate energy transferred in a reaction and hence the enthalpy change of the reaction. This will be limited to experiments where substances are mixed in an insulated container and combustion experiments

e recall Hess's Law and apply it to calculating enthalpy changes of reaction from data provided, selected from a table of data or obtained from experiments and understand why standard data is necessary to carry out calculations of this type

f evaluate the results obtained from experiments using the expression:

*energy transferred in joules = mass × specific heat capacity × temperature change*

and comment on sources of error and assumptions made in the experiments. The following types of experiments should be performed:

i experiments in which substances are mixed in an insulated container and the temperature rise measured

ii simple enthalpy of combustion experiments using, e.g. a series of alcohols in a spirit burner

iii plan and carry out an experiment where the enthalpy change cannot be measured directly, e.g. the enthalpy change for the decomposition of calcium carbonate using the enthalpy changes of reaction of calcium carbonate and calcium oxide with hydrochloric acid

g demonstrate an understanding of the terms *bond enthalpy* and *mean bond enthalpy*, and use bond enthalpies in Hess cycle calculations and recognise their limitations. Understand that bond enthalpy data gives some indication about which bond will break first in a reaction, how easy or difficult it is and therefore how rapidly a reaction will take place at room temperature.